

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-5955
Project Title	PocketSmart AI - Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) *Report*

The proposal report aims to transform budget planning using Generative AI, boosting efficiency and personalized shopping. It tackles overspending and time waste, promising smart budget allocation and curated recommendations. Key features include Gemini 1.5 Flash Pro multimodal AI and cross-platform integration.

Project Overview

Objective	The primary objective is to revolutionize budget planning for home interiors, party events, and jewelry shopping by implementing Gemini 1.5 Flash Pro AI, ensuring personalized and budget-aware recommendations.
Scope	The project comprehensively assesses user budgets, preferences, and images across platforms like Amazon, IKEA, Flipkart, Swiggy, Zomato, OYO for efficient planning.
Problem Statement Description	Addressing overwhelming product choices, budget overruns, and scattered data across e-commerce platforms adversely affects user satisfaction and financial planning.
Impact	Solving these will result in improved budget control, reduced time waste, and enhanced shopping experience contributing to user satisfaction.
Proposed Solution Approach	Employing Gemini 1.5 Flash Pro GenAI to analyze budgets and generate proportional recommendations across Home/Party/Jewelry via Flask/FastAPI.
Key Features	Implementation of multimodal AI recommendation model with routes <code>/generate-home</code> , <code>/generate-party</code> , <code>/generate-jewelry</code> .

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Key Features (continued)

Feature	Description
Real-time personalized suggestions within budget	Dynamically recommends purchases and budget-friendly swaps based on user budget and preferences, updating suggestions instantly as users adjust parameters.
Image upload for jewelry matching	Users can upload photos of jewelry items; the system analyzes style, color, and design to recommend matching clothing, accessories, or similar products within budget.
History page /history and budget breakdown charts	Displays past recommendation history with timesamobs; includes visual charts for budget allocation across categories (Clothing, Accessories, Food, etc.) using interactive bar/pie charts.
PDF export	Export full plan and recommendations as a downloadable PDF report for offline review and sharing, including budget summary and product links.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware — Computing Resources	CPU/GPU, number of cores	T4 GPU / Cloud VM
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, logs	1 TB SSD
Software — Frameworks	Python frameworks	Flask, FastAPI
Libraries	Additional libraries	google-generativeai, Pillow, Flask-Session, CORS, python-dotenv
Development Environment	IDE	VS Code, Jupyter Notebook
Data	Source, size, format	Gemini 1.5 Flash Pro API (text+image), Amazon Product API, IKEA API, Flipkart A(Swiggy/Zomato APIs, OYO API, User budget inputs (JSON, JPG/PNG <5MB)